Jeff Long

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Software Engineer with 15+ years of experience building backend services, distributed systems, and production applications across financial services and healthcare. Specializes in TypeScript, Python, and AWS, with experience developing reusable platform architecture adopted across engineering teams and AI-powered workflows that improved delivery speed by ~33%. Led cross-team architecture initiatives, mentored senior engineers, and owned systems from design through production operations.

Experience

- **Capital One**, Cambridge, MA — August 2022 – September 2026
  Principal Associate Software Engineer
  - Led modernization of legacy infrastructure into independently deployable AWS services using Python, TypeScript, Lambda, Fargate, and DynamoDB, establishing reusable architectural patterns adopted by four engineering teams.
  - Designed two-region event pipelines adopted by dozens of Capital One teams, incorporating retries, dead-letter queues, and idempotency; validated performance and failure recovery with JMeter.
  - Built Claude- and Windsurf-based engineering agents used by eight engineers across two repositories to generate code and tests from reusable OpenSpec specifications, improving delivery speed by ~33% over manual coding.
  - Implemented automated verification pipelines for AI-generated code, integrating unit and end-to-end tests, coverage checks, linting, and specification conformance to validate changes before review.
  - Configured automated AWS deployments using Bogiefile (YAML), internal IaC tooling, Jenkins, and AWS CodeDeploy; implemented least-privilege IAM and migrated secrets for 15+ services from Vault to AWS Secrets Manager.
  - Drove cross-team architectural standards through design reviews and technical guidance, mentoring 10 engineers, including senior engineers, on system design and production reliability.
- **PatientKeeper**, Waltham, MA — 2019-2022
  Senior Software Engineer
  - Built a containerized Python/Flask NLP application that extracted diagnoses and charges from unstructured clinical text into structured data, supporting clinical review workflows in a HIPAA-compliant environment.
  - Modernized physician-facing applications with reusable Vue components and automated tests, integrating user-facing workflows with Java- and Oracle-backed services.
- **Rocket Software**, Waltham, MA — 2018-2019
  Senior Full-Stack Engineer
  - Led replacement of legacy mainframe applications with React/Angular frontends and TypeScript/Node.js services, preserving complex business rules throughout migration.
- **Blue-Soho**, Boston, MA — 2016-2018
  Lead Web Developer
  - Led a four-engineer team delivering web applications for Fortune 500 clients, translating product and design requirements into responsive interfaces while owning architecture and production releases.
- **Rovi**, Boston, MA — 2014-2016
  Software Design Engineer in Test (SDET)
  - Built automated test infrastructure for web applications and data pipelines, covering REST APIs, ETL workflows, Redshift and MySQL data, and CI/CD.

Skills

Languages & Backend: TypeScript, JavaScript, Python, Java, Node.js, REST APIs, SQL, Scala, Go (Golang)

Platform & Architecture: Distributed systems, event-driven architecture, asynchronous workflows, reusable services, system integration, Active Directory/SSO

Cloud & Infrastructure: AWS Lambda, Fargate, IAM, Secrets Manager, S3, DynamoDB, SQS, SNS, Docker, Infrastructure as Code, Jenkins, AWS CodeDeploy, CI/CD

AI & Developer Productivity: Claude, Devin Desktop (formerly Windsurf), OpenSpec, LLM-powered agents, specification-driven development, structured outputs, test generation, automated verification

Frontend: React/Redux, Vue, Angular, Sass, CSS

Data & Persistence: MySQL, Oracle, Redshift, Databricks, ETL

Quality & Production Engineering: Unit testing, Cucumber, Playwright, Automated testing, observability, incident response, JMeter, Splunk, PagerDuty, New Relic

Certifications

- AWS Certified Solutions Architect – Associate
- AWS Certified Machine Learning Engineer – Associate

Education

- **Boston University**, M.S. Computer Systems Engineering
- **Rensselaer Polytechnic Institute**, B.S. Electrical/Computer Systems Engineering